

Multi-Tenant LoRA: Attention Is All You Bleed

Cross-Tenant Data Leakage in Multi-Tenant LoRA Deployments
and Cryptographic Dimension Partitioning as a Structural Remedy

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Abstract

Low-Rank Adaptation (LoRA) is the dominant method for efficient concurrent multi-tenant fine-tuning of large language models, yet it provides *no known* formal guarantees of tenant data isolation.

Multi-tenant LoRA faces Schrödinger’s JIT Swap: merged backbones enable efficiency but leak data, while JIT adapter swapping prevents leakage but eliminates cost savings.

We present six independent empirical measurements demonstrating that *merged-backbone* LoRA deployments leak cross-tenant information through the shared backbone’s attention mechanism. The measurements span multiple granularities: a 2.20% aggregate “commingling premium” above the frozen-backbone information-theoretic ceiling (established via an 8-architecture union test on Mistral-7B), directional cross-tenant accuracy improvement of +1.22%, membership inference at 0.609 AUC on medical data (HIPAA-relevant), 80.8% token-level output divergence from cross-tenant backbone contamination, and direct factual recall of fictional tenant secrets at 20% exfiltration rate. We prove that the root cause is structural: the self-attention dot product $\mathbf{QK}^\top$ is a full inner product across all dimensions, making tenant-specific modifications to attention projections fundamentally unpartitionable.

As a constructive remedy, we introduce *Cryptographic Dimension Partitioning* (CDP), a mechanism that uses HMAC-SHA256-derived binary gate masks to partition a neural network’s hidden layer into disjoint coordinate subspaces, each accessible only with the correct cryptographic key. CDP gates the feed-forward network (FFN) layers—where knowledge is deposited and where element-wise independence permits exact isolation—while leaving the attention mechanism shared. We verify CDP’s core isolation properties in Lean 4 (zero **sorry**, zero new axioms), including gradient isolation, weight confinement, transformer compositionality, and algebraic annihilation. Empirical validation across three model scales (DistilBERT 66M, TinyLlama 1.1B, Mistral-7B) demonstrates accuracy parity with separately trained models ($p > 0.05$, paired t -test) at consolidation ratios up to $R=96$ under co-training, with perfect cross-regime isolation (below-chance accuracy under wrong key) at every regime count tested, including $R=512$.

All proof-of-concept code, Lean 4 proofs, and benchmark scripts are publicly available.¹

Keywords: multi-tenant LLM security, LoRA, low-rank adaptation, tenant isolation, cross-tenant leakage, membership inference, cryptographic partitioning, formal verification, Lean 4

1 Introduction

Low-Rank Adaptation [Hu et al., 2022] has become the *de facto* method for fine-tuning large language models—and increasingly, the preferred *architecture* for deploying them at scale. By freezing a pre-trained backbone and learning low-rank perturbations $\Delta\mathbf{W} = \mathbf{B}\mathbf{A}$ to selected

¹Repository: <https://github.com/ryanmccormick/schemen> Proofs: `proofs/*.lean`

weight matrices, LoRA enforces a clean separation of concerns: the backbone encodes general-purpose capabilities; each adapter encodes domain-specific or tenant-specific knowledge. As the number of foundation models grows and the demand for specialized deployments scales alongside them, this pattern—few backbones, many adapters—is the natural way to structure model portfolios. It is a design choice with tradeoffs, but the modularity is well-motivated.

Multi-tenant serving frameworks—S-LoRA [Sheng et al., 2023], vLLM [Kwon et al., 2023], Punica [Chen et al., 2023]—operationalize this pattern by hosting per-tenant adapters on shared backbone infrastructure, enabling hundreds of tenants on a single GPU. The deployment mode determines the security posture. In *merged-backbone* deployments, multiple tenants’ adapters are arithmetically incorporated into the backbone weights ($\mathbf{W} = \mathbf{W}_0 + \sum_t \mathbf{B}_t \mathbf{A}_t$), yielding a single model that serves all tenants concurrently—the highest throughput, but also the most direct channel for cross-tenant information flow. In *adapter-based* deployments, adapters remain separate objects loaded per-request; the backbone is shared but unmodified at inference time. Even adapter-based serving leaves a critical question unresolved: *what is the relationship between an adapter and a tenant?* Cloud providers define “tenant” at the IAM or account boundary, but the atomic unit of isolation is often finer—a law firm’s tenant is a *matter*, a hospital system’s tenant is a *clinical trial* or *patient cohort*. Multiple adapters may serve a single organizational customer, each requiring isolation from the others.

However, this modularity comes with a fundamental tension, which we term *Schrödinger’s JIT Swap*: merged-backbone deployments leak cross-tenant information, while JIT adapter swapping prevents leakage but eliminates the economic advantage that makes multi-tenant serving viable in the first place. No deployment can satisfy both efficiency and isolation simultaneously.

This architecture implicitly promises tenants that their data is isolated. The mental model is one of “database rows”: each adapter is a separate record, and the backbone is shared infrastructure that treats each record independently. In merged-backbone deployments, we demonstrate that this mental model is fundamentally flawed.

LoRA adapters achieve their power by modifying the backbone’s attention projections. The attention mechanism computes $\text{softmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d_k}) \cdot \mathbf{V}$, where $\mathbf{Q} = \mathbf{x}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{x}\mathbf{W}_K$, $\mathbf{V} = \mathbf{x}\mathbf{W}_V$. The critical operation is $\mathbf{Q}\mathbf{K}^\top$: a *full inner product across all d_k dimensions*. When LoRA modifies $\mathbf{W}_Q$ via $\mathbf{W}_Q + \mathbf{B}_Q \mathbf{A}_Q$, it changes every element of the query vectors. Those modified queries are then dotted against keys across *all* dimensions. Information cannot be partitioned through a dot product—the dot product is the mathematical operation that combines information from every dimension into a single scalar. This is not incidental. This is the entire point of attention.

The consequence is that every LoRA adapter that modifies attention projections using tenant-specific data creates information in the shared backbone that is accessible to all other tenants. We call the measurable manifestation of this the *commingling premium*: the accuracy that LoRA achieves above what any classifier—regardless of architecture or capacity—can extract from the frozen backbone. Our union test (Section 5) establishes this ceiling at 91.89% using eight classifier architectures up to 26 million parameters on Mistral-7B. LoRA rank-4, with only 400K parameters, reaches 94.09%—exceeding the ceiling by 2.20 percentage points. The extra accuracy is information that *does not exist* in the frozen representation. It was created by LoRA modifying the backbone using tenant training data.

We present six increasingly direct measurements of this leakage for *merged-backbone* deployments (Section 6), ranging from aggregate accuracy shifts to literal factual recall of fictional secrets across tenant boundaries. We corroborate our findings with independent academic validation: Ran et al. [Ran et al., 2025] demonstrated 0.775 AUC membership inference against LoRA-tuned models using a pre-trained reference attack, and a 10-paper literature review spanning NeurIPS 2024 through NDSS 2026 confirms that no existing PEFT method provides formal multi-tenant isolation.

As a constructive contribution, we introduce Cryptographic Dimension Partitioning (CDP,

Section 7).² CDP uses HMAC-SHA256 in counter mode to drive a Fisher-Yates shuffle that deterministically partitions a hidden layer’s dimensions into R non-overlapping regimes. A binary gate mask—1s for owned dimensions, 0s elsewhere—is applied element-wise after the FFN activation during both training and inference. During backpropagation, the mask zeroes gradients for excluded dimensions, confining knowledge deposition to the regime’s partition.

The key architectural insight is that FFN layers are element-wise: $h_j = \sigma(\mathbf{x} \cdot \mathbf{W}_1[:, j] + b_j)$. Zeroing h_j via the gate eliminates dimension j ’s contribution to the output exactly. No other dimension is affected. This is why FFN gating provides provable isolation while attention modification cannot be partitioned—the inner product entangles all dimensions before any gate could act.

We verify CDP’s properties in Lean 4 (Section 9). The core security chain—gradient isolation, weight confinement for both $\mathbf{W}_1$ and $\mathbf{W}_2$, transformer block compositionality, combinatorial hardness of key recovery, steganographic failure under wrong-key access, and algebraic annihilation upon regime destruction—is fully machine-checked with zero `sorry` and zero new axioms.

Contributions.

1. **The frozen-backbone ceiling methodology.** We introduce the union test: train the most powerful classifier achievable on frozen backbone features, with all dimensions, unlimited architecture, and full training budget. Any accuracy above this ceiling requires backbone modification—and in multi-tenant LoRA, backbone modification is cross-tenant information flow.
2. **Six independent leakage measurements** spanning aggregate commingling (2.20%), directional cross-tenant flow (+1.22%), membership inference on medical data (0.609 AUC), token-level output corruption (80.8%), and factual knowledge exfiltration (20% recall of fictional secrets).
3. **Cryptographic Dimension Partitioning (CDP)**, a mechanism that provides provable tenant isolation in multi-tenant neural networks by gating FFN layers with cryptographically derived binary masks.
4. **Machine-checked Lean 4 proofs** formalizing CDP’s isolation, confinement, and destruction properties (zero `sorry`, zero new axioms).
5. **Empirical validation at three model scales** (66M, 1.1B, 7B parameters) demonstrating accuracy parity with separate models and perfect cross-regime isolation at regime counts up to $R=512$.

2 Background and Related Work

2.1 Low-Rank Adaptation (LoRA)

LoRA [Hu et al., 2022] freezes a pre-trained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$ and learns a low-rank decomposition $\Delta \mathbf{W} = \mathbf{B}\mathbf{A}$ where $\mathbf{B} \in \mathbb{R}^{d \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. The effective weight at inference is $\mathbf{W} = \mathbf{W}_0 + \mathbf{B}\mathbf{A}$. By targeting attention projections ($\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V, \mathbf{W}_O$), LoRA achieves competitive fine-tuning quality with 0.1–1% of full-model parameters. QLoRA [Detrmers et al., 2024] extends this to 4-bit quantized backbones.

2.2 Multi-Tenant LoRA Serving

S-LoRA [Sheng et al., 2023] introduces unified memory management for thousands of concurrent LoRA adapters, enabling batched multi-adapter serving on shared GPUs. vLLM [Kwon et al., 2023] implements PagedAttention with native multi-LoRA support. Punica [Chen et al., 2023] provides custom CUDA kernels for batched adapter computation. All three frameworks

²Proof-of-concept code and Lean 4 proofs: <https://github.com/ryanmccormick/schemen>

support both adapter-based and merged-backbone modes (Section 1); it is the merged-backbone configuration that enables the leakage we measure.

2.3 Membership Inference Against LoRA

Ran et al. [Ran et al., 2025] demonstrated membership inference attacks against LoRA-tuned models at 0.775 AUC by comparing the fine-tuned model’s behavior to the publicly available pre-trained model. Their strongest finding: the *existence* of the pre-trained reference model makes LoRA more vulnerable than full fine-tuning to membership inference, because it provides a perfect “before” snapshot. The only effective defense (DP-SGD) incurs $31\times$ training overhead with severe utility degradation.

SPV-MIA [Fu et al., 2024] achieves ~ 0.9 AUC via self-prompt calibration. ICP-MIA [Lu et al., 2026] reaches 0.802–0.829 AUC specifically on LoRA configurations (rank 32–64) and shows that higher LoRA rank *increases* vulnerability. The literature consensus: LoRA reduces extractive memorization (verbatim reproduction) but remains highly vulnerable to membership inference [Wang and Li, 2025, Das et al., 2025, Mireshghallah et al., 2022].

2.4 OWASP LLM Top 10 (2025)

OWASP’s Top 10 for Large Language Model Applications [OWASP, 2025] independently cataloged the vulnerabilities that motivate this work. **LLM08:2025** (Vector and Embedding Weaknesses) identifies cross-context information leaks between tenants sharing the same vector space and prescribes “strict logical and access partitioning” as the mitigation. Our finding that the attention inner product is structurally unpartitionable means LoRA cannot satisfy this requirement—the prescribed mitigation is only achievable via a mechanism like CDP. **LLM03:2025** identifies LoRA adapters as a supply chain risk.

2.5 Superposition in Neural Networks

Elhage et al. [Elhage et al., 2022] demonstrated that neural networks represent more features than they have dimensions via *superposition*: features are encoded as nearly-orthogonal directions in a shared space. This finding is foundational to our work. CDP’s gate *coerces* superposition: the backbone is forced to encode features that survive arbitrary coordinate projections (the gate mask), distributing features redundantly so that any sufficiently large coordinate subset can decode them (Section 8).

3 Threat Model

We consider a standard *merged-backbone* multi-tenant LoRA deployment where N tenants share a pre-trained backbone $\mathbf{W}_0$. Each tenant i trains a LoRA adapter $(\mathbf{B}_i, \mathbf{A}_i)$ on private data $\mathcal{D}_i$. In this setting, adapters are permanently incorporated into the backbone weights $(\mathbf{W}' = \mathbf{W}_0 + \sum_i \mathbf{B}_i \mathbf{A}_i)$, enabling concurrent serving within the same batch [Sheng et al., 2023].

Note on serving architectures. Our threat model and all leakage measurements assume *merged-backbone* deployments. In JIT-adapter-swapping architectures (where adapters are loaded/unloaded per request without merging), the leakage mechanism differs (e.g., shared KV caches). We discuss this distinction in the limitations (Section 13) and develop the broader efficiency-isolation tradeoff in Section 14.1.

Adversary model. The adversary is a co-tenant with legitimate API access. They can submit inference queries and observe outputs (tokens, logits, or embeddings). They cannot inspect model weights directly. This is the weakest possible adversary—the “attack” is running a prompt.

Violation. Cross-tenant information flow occurs when Tenant j 's inference results are influenced by Tenant i 's training data $\mathcal{D}_i$, where $i \neq j$. We measure this at five granularities:

1. **Aggregate:** Tenant j 's downstream accuracy improves after Tenant i 's adapter modifies the backbone.
2. **Directional:** The improvement is measurable, reproducible, and directional ($A \rightarrow B$, not noise).
3. **Per-sample:** An adversary can identify whether a specific sample was in $\mathcal{D}_i$ (membership inference).
4. **Token-level:** Tenant j 's generated tokens are altered by the backbone modifications from $\mathcal{D}_i$.
5. **Factual:** Tenant j 's model can answer questions about facts in $\mathcal{D}_i$.

4 Why Self-Attention Cannot Be Partitioned

We now establish the theoretical foundation for why LoRA's cross-tenant leakage is structural, not incidental.

Definition 1 (Self-Attention). *For input $\mathbf{x} \in \mathbb{R}^{n \times d}$, weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d_k}$, and output projection $\mathbf{W}_O \in \mathbb{R}^{d_k \times d}$:*

$$\mathbf{Q} = \mathbf{x}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{x}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{x}\mathbf{W}_V \quad (1)$$

$$\text{Attn}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}\mathbf{W}_O \quad (2)$$

Proposition 1 (Attention Entanglement). *Let $\mathbf{W}'_Q = \mathbf{W}_Q + \mathbf{B}_Q\mathbf{A}_Q$ be a LoRA modification to the query projection, where $\mathbf{B}_Q \in \mathbb{R}^{d \times r}$ and $\mathbf{A}_Q \in \mathbb{R}^{r \times d_k}$ were trained on tenant data $\mathcal{D}_i$. For any partition of dimensions $\{1, \dots, d_k\} = S_i \sqcup S_j$ (where $S_i \cap S_j = \emptyset$), the modified attention scores $\mathbf{Q}'\mathbf{K}^\top$ cannot be decomposed into tenant-isolated components:*

$$[\mathbf{Q}'\mathbf{K}^\top]_{ab} = \sum_{l=1}^{d_k} Q'_{al} K_{bl}$$

Every dimension l contributes to every attention score. A LoRA modification to $\mathbf{W}_Q$ that changes Q'_{al} at any dimension propagates to all attention scores involving position a .

Proof. The (a, b) entry of $\mathbf{Q}'\mathbf{K}^\top$ is $\sum_{l=1}^{d_k} Q'_{al} K_{bl} = \sum_l (\mathbf{x}_a(\mathbf{W}_Q + \mathbf{B}_Q\mathbf{A}_Q))_l \cdot K_{bl}$. The LoRA perturbation $\mathbf{B}_Q\mathbf{A}_Q$ is a dense rank- r matrix. For any $l \in \{1, \dots, d_k\}$, the perturbed query element Q'_{al} depends on $\mathbf{x}_a\mathbf{B}_Q\mathbf{A}_Q[:, l]$, which involves *all* input dimensions through $\mathbf{x}_a\mathbf{B}_Q$. Each attention score is therefore a function of the full LoRA perturbation, with no dimension-wise decomposability. $\square$

Corollary 2 (LoRA Multi-Tenant Commingling). *In any transformer using self-attention, LoRA modifications to attention projections trained on $\mathcal{D}_i$ influence the attention computation for all inputs, including those from tenant $j \neq i$. No dimension-based partition can isolate the effect.*

This holds universally for every transformer architecture (GPT, LLaMA, Mistral, Falcon, T5, BERT) because self-attention is universal in modern LLMs.

Definition 2 (FFN Element-Wise Independence). *In a standard feed-forward network layer:*

$$\text{FFN}(\mathbf{x}) = \sigma(\mathbf{x}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2$$

Each hidden dimension j is computed independently: $h_j = \sigma(\mathbf{x} \cdot \mathbf{W}_1[:, j] + b_{1,j})$. Zeroing h_j eliminates dimension j 's contribution to $\text{FFN}(\mathbf{x})$ exactly, with no effect on any other dimension $j' \neq j$.

This element-wise independence is why FFN layers can be partitioned while attention cannot. CDP exploits this structural asymmetry.

4.1 Three Vectors of Cross-Tenant Information Flow

The structural leakage manifests through three independent mechanisms:

1. **Merged-backbone contamination.** When adapters are permanently incorporated into the backbone ($\mathbf{W}' = \mathbf{W}_0 + \mathbf{BA}$), the weight perturbation reshapes the backbone’s geometry for all subsequent computation. Measurements 4 and 5 demonstrate this vector.
2. **Co-training interference.** When multiple adapters train concurrently on a shared backbone, each adapter’s gradient updates influence the shared attention representations. Even without permanent merging, the shared forward pass during batched training creates cross-tenant signal.
3. **Rank-proportional capacity leakage.** The adapter’s rank r determines how much tenant-specific information is encoded in $\Delta\mathbf{W} = \mathbf{BA}$. Higher rank provides more capacity for encoding training data structure—and more surface for membership inference. ICP-MIA [Lu et al., 2026] demonstrated this directly: MIA AUC increases monotonically with LoRA rank (0.802 at rank 32 to 0.829 at rank 64).

These three vectors are independent and cumulative. No single mitigation addresses all three.

5 The Frozen-Backbone Ceiling (Union Test)

5.1 Methodology

The union test establishes the information-theoretic upper bound on classification accuracy achievable from a frozen backbone’s representations. We extract last-token hidden states from Mistral-7B v0.3 (4-bit quantized, 4096 dimensions) on AG News 4-class classification (20,000 train / 7,600 test), then train the most powerful post-hoc classifier achievable:

- All 4096 dimensions (no gating, no isolation)
- Eight classifier architectures (linear probe through 3-hidden-layer MLPs up to 26M parameters)
- Full training budget (30 epochs, 3 seeds: 42, 123, 456)

If *any* frozen-backbone classifier matches LoRA, then LoRA’s advantage is explainable without backbone modification. If none can, the backbone modification is proven.

5.2 Results

Table 1: Union test results: Frozen-backbone classifiers vs. LoRA (Mistral-7B, AG News).

Architecture	Params	Accuracy	vs. LoRA
Linear probe (4096 $\rightarrow$ 4)	16K	90.24%	−3.85%
MLP (512)	2.1M	91.56%	−2.53%
MLP (512, 256)	2.2M	91.89%	−2.20%
MLP (1024, 512)	4.7M	91.77%	−2.32%
MLP (2048, 1024, 512)	11.0M	91.43%	−2.66%
MLP (4096, 2048, 512)	26.2M	91.75%	−2.34%
LoRA rank-4 (Q+V)	400K	94.09%	—

The frozen ceiling is 91.89%. Five MLP architectures (2.1M to 26.2M parameters) converge to 91.4–91.9%. The 26M-parameter model scores *lower* than the 2.2M model—more capacity does not help, because the information is not in the frozen representation. LoRA rank-4, with $5\times$ fewer parameters than the best MLP and $65\times$ fewer than the largest, exceeds the ceiling by 2.20 percentage points.

Definition 3 (Commingling Premium). *The commingling premium δ is the accuracy gap between a LoRA-adapted model and the frozen-backbone ceiling:*

$$\delta = \text{Acc}_{\text{LoRA}} - \max_{\theta} \text{Acc}(\theta; \text{frozen features})$$

where the maximum is taken over all post-hoc classifier architectures θ . A positive δ requires backbone modification—information that does not exist in the frozen representation.

The measured commingling premium is $\delta = 2.20\%$. In multi-tenant deployment, this backbone modification is performed using tenant-specific data and is accessible to all co-tenants.

The measurement is binary, not graduated. Any positive δ —regardless of magnitude—constitutes proof that tenant-specific training data influenced the backbone’s representational geometry. A commingling premium of 0.001% and one of 20% carry the same security conclusion: information crossed a trust boundary without authorization. In security engineering, a lock that opens to the wrong key 0.1% of the time is a broken lock. The frozen-backbone ceiling establishes a hard floor; any accuracy above it is unauthorized cross-tenant information flow. The numbers that follow should be read as *binary evidence of violation*, not as magnitudes to be weighed against a threshold of acceptability.

6 Experimental Evidence of Cross-Tenant Leakage

We present six independent measurements at increasing levels of directness. All experiments use Mistral-7B v0.3 (4-bit NF4 quantization) unless otherwise noted. Full experimental configurations, including hyperparameters and random seeds, are provided in the appendix.

6.1 Measurement 1: Commingling Premium (Series 5b)

Result: $\delta = 2.20\%$. Established via the union test (Section 5). The frozen-backbone ceiling is 91.89%. LoRA rank-4 reaches 94.09%. The 2.20% gap is information created by modifying the backbone’s attention computation using tenant training data.

6.2 Measurement 2: Directional Cross-Tenant Flow (Series 5c)

Result: $+1.22\%$ cross-tenant accuracy improvement.

AG News is split into two tenants with non-overlapping label spaces: Tenant A (World, Sports) and Tenant B (Business, Sci/Tech). Phase 1 trains linear probes on frozen features (baseline). Phase 2 fine-tunes LoRA using *only* Tenant A’s data. Phase 3 re-extracts features from the LoRA-modified backbone and re-probes Tenant B.

Table 2: Cross-tenant accuracy improvement after Tenant A’s LoRA training (3-seed mean).

	Frozen Baseline	After A’s LoRA	Δ
Tenant A (trained)	98.57%	99.42%	+0.85%
Tenant B (untrained)	88.89%	90.11%	+1.22%

Tenant B’s accuracy improves by 1.22% from training that used *only* Tenant A’s data. The two tenants have disjoint label spaces. The improvement is reproducible across seeds ($\sigma < 0.3\%$). This is directional, measurable cross-tenant information flow.

As a control, a CDP gate partition on the same frozen features shows zero cross-regime benefit: using Tenant A’s gate mask on Tenant B’s features produces below-chance accuracy (21–22% vs. 25% chance).

6.3 Measurement 3: Membership Inference on Medical Data (Series 5d)

Result: 0.609 AUC on HIPAA-relevant medical data.

We reproduce the LoRA-Leak methodology [Ran et al., 2025] on medical data. LoRA (rank 8, alpha 16, 3 epochs) is trained on Qwen2.5-1.5B using 5,089 MedQA samples (USMLE-style medical questions). Three MIA score functions are evaluated:

Table 3: Membership inference AUC: LoRA vs. gated-FFN control (3-seed mean, $\sigma < 0.001$).

Score Function	LoRA AUC	Gated Control AUC
LOSS+Pre	0.609	0.499 (random)
Ratio	0.609	0.500 (random)
Min-K%+Pre	0.501	0.498 (random)

An adversary with API access to a LoRA fine-tuned medical model can determine—with 60.9% reliability—which specific clinical scenarios were in a healthcare organization’s training data. The gated-FFN control produces AUC 0.499 (perfectly random), because the backbone is bit-identical to the public pre-trained model, leaving no “before vs. after” signal for the attacker.

6.4 Measurement 4: Token-Level Output Divergence (Series 5e)

Result: 80.8% token divergence across domains.

“Capture the flag” protocol: Tenant A (medical, MedQA) and Tenant B (legal, contract law) train on Qwen2.5-1.5B. We compare Tenant B’s generated responses on a clean backbone (no Tenant A) vs. a contaminated backbone (after Tenant A’s LoRA is merged).

Table 4: Token-level divergence from cross-tenant backbone contamination.

Prompt Category	Token Divergence	KL Divergence (nats)
Medical prompts	77.0%	15.03
Legal prompts	85.3%	16.95
Neutral prompts	79.4%	12.96
Overall	80.8%	15.38

Four out of every five tokens in Tenant B’s legal responses are different because a medical organization trained on the same backbone. The contamination manifests as distributional shift—altered phrasing, structure, and synonym choice—not as domain-specific vocabulary leakage.

6.5 Measurement 5: Factual Knowledge Exfiltration (Series 5f)

Result: 6/30 fictional secrets recalled by the wrong tenant.

We fabricate 30 confidential facts across legal, financial, and protocol domains using fictional company names and invented numbers that exist nowhere on the internet. Tenant B’s LoRA is trained on these facts, merged into the backbone. Tenant A (medical) then trains on the contaminated backbone. We probe Tenant A with Tenant B’s questions. Binary scoring: does the response contain the exact correct number or entity?

Examples of exfiltrated secrets: “Orion Vanguard arbitration location: Stockholm,” “Meridian Dynamics Series C valuation: \$520M,” “Pyralis deployment F1 threshold: 0.92.” The medical model was never trained on these facts. It learned them through the shared backbone.

The mechanism: LoRA encodes a low-rank projection of the relational structure of its training corpus—the “statistical shadow” of the data. When merged into the backbone ($\mathbf{W}' = \mathbf{W} + \mathbf{BA}$),

Table 5: Cross-tenant knowledge exfiltration (fictional secrets).

Phase	Secrets Recalled
Pre-trained baseline	0/30
Tenant B self-test	30/30
Tenant A probed with B’s secrets	6/30 (20%)
CDP gated control	0/30

those relational paths become part of the backbone’s weight geometry and persist when adapters are swapped.

6.6 Measurement 6: Independent Academic Validation

Ran et al. [Ran et al., 2025] independently demonstrated 0.775 AUC membership inference on LoRA fine-tuned models (GPT-2 XL, Pythia-2.8B, Llama-2 7B) including on AG News with Llama-2 (0.765 AUC). Their finding that the only effective defense (DP-SGD) incurs catastrophic utility loss corroborates our conclusion that the leakage is structural, not a hyperparameter issue.

6.7 Summary of Measurements

Table 6: Six independent measurements of LoRA cross-tenant leakage.

#	Measurement	Result	Granularity
1	Commingle premium	2.20% above ceiling	Aggregate
2	Directional flow	+1.22% (A→B)	Directional
3	Medical MIA	0.609 AUC	Per-sample
4	Token divergence	80.8%	Per-token
5	Knowledge exfiltration	6/30 (20%)	Per-fact
6	Academic (LoRA-Leak)	0.775 AUC	Per-sample

7 Cryptographic Dimension Partitioning

7.1 Overview

Cryptographic Dimension Partitioning (CDP) uses cryptographic key material to partition a neural network’s hidden layer into R disjoint coordinate subspaces (“regimes”). Each regime is accessible only with the correct key. The partition is enforced by a binary gate mask applied at the FFN layer—where element-wise independence (Definition 2) permits exact isolation.

7.2 Gate Mask Derivation

The Fisher-Yates shuffle backed by HMAC-SHA256 in counter mode produces a cryptographically uniform permutation. Rejection sampling eliminates modulo bias (proven: `rejection_sampling_count` in Lean 4). The resulting masks satisfy:

$$\mathbf{g}_r \odot \mathbf{g}_s = \mathbf{0} \quad \forall r \neq s, \quad \sum_{r=0}^{R-1} \mathbf{g}_r = \mathbf{1} \quad (3)$$

where $\odot$ denotes element-wise multiplication. The masks are disjoint and exhaustive.

Algorithm 1 Gate mask derivation from cryptographic key

Require: Master key $k \in \{0, 1\}^{256}$, number of regimes R , hidden dimension d

Ensure: R disjoint binary masks $\{\mathbf{g}_0, \dots, \mathbf{g}_{R-1}\}$ with $\sum_r \mathbf{g}_r = \mathbf{1}$

```
1: Initialize index array  $\pi = [0, 1, \dots, d-1]$ 
2: for  $i = d-1$  down to 1 do ▷ Fisher-Yates shuffle
3:    $\text{ctr} \leftarrow \text{HMAC-SHA256}(k, i)$ 
4:    $j \leftarrow \text{ctr} \bmod (i+1)$  ▷ Rejection sampling: zero modulo bias
5:   Swap  $\pi[i] \leftrightarrow \pi[j]$ 
6: end for
7: Partition  $\pi$  into  $R$  contiguous blocks of size  $\lfloor d/R \rfloor$  (or  $\lceil d/R \rceil$ )
8: for  $r = 0$  to  $R-1$  do
9:    $\mathbf{g}_r \in \{0, 1\}^d$  with  $g_{r,j} = 1$  iff  $j \in \text{block}_r(\pi)$ 
10: end for
```

7.3 Gated Forward Pass

For an input $\mathbf{x}$, regime r with gate mask $\mathbf{g}_r$:

$$\mathbf{h} = \sigma(\mathbf{x}\mathbf{W}_1 + \mathbf{b}_1) \quad (4)$$

$$\tilde{\mathbf{h}} = \mathbf{h} \odot \mathbf{g}_r \quad (5)$$

$$\hat{\mathbf{y}} = \tilde{\mathbf{h}}\mathbf{W}_2 + \mathbf{b}_2 \quad (6)$$

The gate mask (5) zeroes all dimensions outside regime r 's partition. The output $\hat{\mathbf{y}}$ depends only on the $\lfloor d/R \rfloor$ dimensions owned by regime r .

7.4 Gradient Isolation

During backpropagation, the gate mask confines gradients:

$$\frac{\partial \mathcal{L}}{\partial \tilde{\mathbf{h}}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \mathbf{W}_2^\top \quad (7)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}} = \frac{\partial \mathcal{L}}{\partial \tilde{\mathbf{h}}} \odot \mathbf{g}_r \quad (8)$$

For any dimension $j \notin \text{supp}(\mathbf{g}_r)$: $g_{r,j} = 0 \implies \frac{\partial \mathcal{L}}{\partial h_j} = 0$. The gradient is exactly zero, not approximately zero. Consequently:

- $\mathbf{W}_1[:, j]$ receives zero gradient update for $j \notin \text{supp}(\mathbf{g}_r)$.
- $\mathbf{W}_2[j, :]$ receives zero gradient update for $j \notin \text{supp}(\mathbf{g}_r)$ because $\tilde{h}_j = 0$.

Knowledge deposition is confined to the regime's partition during training. This is the *dual* of the forward isolation: the gate restricts both what the output can source (forward) and where gradients can flow (backward).

Isolation enables consent. CDP does not merely prevent cross-tenant information flow—it makes the *default* state isolated and any sharing *explicit*. Two regimes that wish to pool inference can be granted overlapping or shared key material under contract; the sharing is auditable, revocable, and cryptographically scoped. Without a mechanism like CDP, there is no way to opt *out* of cross-tenant flow, and therefore no meaningful way to opt *in*. Consent requires that isolation is the default from which sharing is a deliberate departure.

7.5 Key Hierarchy and Token System

CDP uses HKDF-Expand (RFC 5869) with HMAC-SHA256 to derive a strict one-way key hierarchy:

$$\text{MasterKey} \xrightarrow{\text{HKDF}} \text{RegimeKey}_r \xrightarrow{\text{HKDF}} \text{TenantKey}_{r,t}$$

Mask tokens are distributed via AES-256-GCM with all metadata (tenant ID, regime ID, dimensions, version) bound into the authentication tag as Associated Authenticated Data (AAD). Tampering with any field causes decryption failure.

7.6 Regime Destruction (Algebraic Annihilation)

Destroying regime r consists of zeroing the weight columns and rows corresponding to $\text{supp}(\mathbf{g}_r)$:

$$\mathbf{W}_1[:, j] \leftarrow \mathbf{0}, \quad \mathbf{W}_2[j, :] \leftarrow \mathbf{0} \quad \forall j \in \text{supp}(\mathbf{g}_r) \quad (9)$$

By the confinement theorems, no knowledge from regime r exists outside $\text{supp}(\mathbf{g}_r)$. After zeroing, the model is provably identical to one that was never trained on regime r ’s data. This is algebraic annihilation: not “file deleted” (recoverable from backups), but mathematical elimination of the learned representation (irreversible).

8 Coerced Superposition

An unexpected finding: the gate does not degrade representational quality as naïvely expected. Halving the available dimensions should, in a linear model, halve the representational capacity. Empirically, accuracy is statistically indistinguishable from separately trained models even at extreme partitioning ($R=96$, 8 dimensions per regime on a 768-dimensional model).

The mechanism is *coerced superposition*. The gate constrains the output geometry: each regime sees only its slice of the hidden dimension. For the backbone to minimize loss across all regimes simultaneously, it must learn representations that survive arbitrary coordinate projections. This forces the backbone to encode more useful feature directions than any single regime has dimensions, distributing features redundantly so that any sufficiently large coordinate subset can decode them.

This was not predicted theoretically. It was discovered empirically through accuracy benchmarks and subsequently explained by connection to the superposition phenomenon documented by Elhage et al. [2022]. The gate doesn’t *allow* superposition—it *coerces* it. The backbone must increase superposition density or suffer quality degradation across all regimes simultaneously.

This coerced superposition produces a structural quality advantage over LoRA. LoRA freezes the backbone by definition: the pre-trained representation geometry is fixed, and every adapter independently compensates for the same frozen limitations via rank-constrained perturbations. CDP co-training lets the backbone reshape itself: attention weights learn routing patterns that serve all regimes, layer norms adapt to the combined distribution, and the backbone actively packs features into coordinate subspaces that survive the gate projection.

The empirical evidence is stark. At 64 dimensions per regime: DistilBERT co-trained shows -0.07% gap (noise). Mistral-7B frozen shows -13.33% gap. Same gating mechanism, same dimension partitioning, $190\times$ worse gap. The only difference is whether the backbone can adapt.

9 Formal Verification in Lean 4

We verify CDP’s isolation properties in Lean 4, organized across fifteen proof files. All proofs are fully verified by the Lean 4 kernel with zero `sorry` (no unfinished proofs) and zero new axioms (no trust assumptions beyond Lean’s kernel).

9.1 Core Theorems

Table 7: Selected key theorems from the Lean 4 proof suite.

Theorem	Statement
<code>gradient_isolation</code>	If $g_{r,j} = 0$, then the gradient with respect to h_j is exactly zero, unconditionally for all losses, optimizers, and inputs.
<code>weight_update_confined</code>	Weight updates to $\mathbf{W}_1$ are confined to columns in $\text{supp}(\mathbf{g}_r)$.
<code>w2_update_confined</code>	Weight updates to $\mathbf{W}_2$ are confined to rows in $\text{supp}(\mathbf{g}_r)$.
<code>masks_orthogonal</code>	For $r \neq s$: $\langle \mathbf{g}_r, \mathbf{g}_s \rangle = 0$ (disjoint supports).
<code>wrong_mask_reads_wrong_dimensions</code>	Applying $\mathbf{g}_s$ to activations trained under $\mathbf{g}_r$ reads dimensions with zero trained signal.
<code>access_requires_exact_mask</code>	Only the exact correct mask reproduces the correct output.
<code>compose_disjoint</code>	Union of disjoint regime masks preserves constituent access.
<code>residual_block_full_confinement</code>	Confinement holds through a complete transformer block (FFN + residual connection).
<code>non_preserving_breaks_confinement</code>	Confinement placed between activation and gate <i>breaks</i> confinement (negative result).
<code>adversary_sees_zero</code>	Under wrong mask, the gated activations are exactly zero at all regime-specific dimensions.
<code>rejection_sampling_count</code>	The HMAC-backed rejection sampling has zero modulo bias.

9.2 Scope and Limitations

The proofs cover gradient isolation, weight confinement (both $\mathbf{W}_1$ and $\mathbf{W}_2$), combinatorial hardness of key recovery, steganographic failure, transformer compositionality (attention + residual + LayerNorm), support preservation under gate application, and compressed extraction equivalence.

The proofs are scoped to the *gated FFN dimensions*. The attention mechanism, token embeddings, and output projection are shared across all regimes and are *not* covered by the isolation guarantees. CDP isolates the fine-tuning delta deposited in FFN layers. The shared backbone layers represent a trust boundary that tenants implicitly accept.

The key security assumption is that HMAC-SHA256 is a pseudorandom function (the standard PRF assumption). If HMAC-SHA256 is broken, the internet’s TLS infrastructure faces larger problems than CDP.

10 Empirical Validation

10.1 Setup

We validate CDP across three model scales:

All experiments use AG News 4-class classification (8,000–20,000 train / 7,600 test). Statistical tests use paired t -tests across 3–5 seeds. Hardware: NVIDIA A100-80GB (Modal cloud), RTX 4090 (local).

Table 8: Validation configurations.

Model	Hidden dim	Params	R tested	Training
DistilBERT-base	768	66M	8–128	Co-trained
TinyLlama	2048	1.1B	4	Co-trained
Mistral-7B v0.3	4096	7B	4–512	Frozen (4-bit)

10.2 Accuracy Parity Under Co-Training (Series 1)

DistilBERT co-trained on 7 parallel A100s, 11.6 minutes wall clock.

Table 9: Co-trained accuracy: gated vs. separate models (DistilBERT, AG News).

R	Dims/Regime	Gated	Separate	Gap
8	96	91.29%	91.22%	+0.07%
16	48	91.12%	91.31%	−0.19%
24	32	91.10%	91.17%	−0.07%
32	24	91.11%	91.02%	+0.09%
64	12	91.03%	91.10%	−0.07%
96	8	90.73%	90.88%	−0.15%
128	6	90.25%	91.12%	−0.87%

R=8 through R=96: viable. Accuracy holds above 90.7% with gaps under 0.2%. Statistical parity (3-seed paired t -test): R=8 $p=0.379$, R=16 $p=0.441$. We cannot reject the null hypothesis that gated = separate. The co-trained backbone has adapted its superposition to sustain all regimes.

R=128 (6 dims/regime): first meaningful degradation (−0.87%). The practical co-trained ceiling for DistilBERT on this task is R=96.

10.3 Cross-Regime Isolation

Table 10: Wrong-mask accuracy at every R tested (DistilBERT and Mistral-7B).

Model	R	Cross-Regime Acc	Chance
DistilBERT	8–128	23.13–23.24%	25%
Mistral-7B	4–512	21.1–23.2%	25%

Wrong-mask accuracy is *below chance* at every R tested, including R=512 on Mistral-7B (8 dims/regime, 46.8% correct-mask accuracy, 23.1% wrong-mask). Isolation never breaks. The below-chance performance is expected: the wrong mask activates dimensions trained for a different label permutation, actively misdirecting the classifier.

10.4 Frozen Backbone Validation (Series 4)

R-sweep on Mistral-7B with frozen 4-bit backbone:

The same gating mechanism that delivers <0.2% gaps on co-trained DistilBERT delivers 2.5–43% gaps on frozen Mistral-7B. At matched dimension count (64 dims/regime): DistilBERT co-trained −0.07%, Mistral frozen −13.33%. 190× worse gap from the same gate architecture. The only difference is whether the backbone can adapt.

Table 11: Frozen-backbone accuracy (Mistral-7B, 4-bit, AG News).

R	Dims/Regime	Accuracy	Gap
4	1024	87.62%	−2.49%
8	512	86.78%	−3.32%
16	256	85.96%	−4.14%
64	64	76.77%	−13.33%
512	8	~46.8%	−43.3%

This frozen-backbone effect applies equally to LoRA: every LoRA adapter operates on a frozen backbone by definition and suffers the same structural limitation.

10.5 Adapter-Gated Architecture (Series 5c)

An additive gated adapter (shared low-rank projection $\rightarrow$ gate $\rightarrow$ head) on frozen Mistral-7B recovers most of the capacity lost to freezing:

Table 12: Adapter-gated architecture on frozen Mistral-7B vs. union ceiling.

R	Best Adapter+MLP	Union Ceiling	Gap
4	91.37%	91.89%	−0.52%
8	91.55%	91.89%	−0.34%
16	91.42%	91.89%	−0.47%

The adapter eliminates R-sensitivity (R=4 to R=16: <0.2% variance) while maintaining proven isolation (cross-regime accuracy: 21–22%, below chance at every R). The gate costs 0.34 percentage points relative to the frozen ceiling. The frozen backbone, not the gate, is the real constraint.

11 OWASP Alignment

OWASP LLM08:2025 (Vector and Embedding Weaknesses) identifies cross-context information leaks between tenants sharing the same vector space and prescribes “strict logical and access partitioning” as the mitigation. Our finding that the attention inner product is structurally unpartitionable (Theorem 1) means LoRA cannot satisfy this requirement. CDP is the only published mechanism that achieves the partitioning OWASP prescribes for multi-tenant model isolation:

12 Comparison with Existing Defenses

The only existing defense that meaningfully reduces MIA on LoRA is differential privacy (DP-SGD), which incurs $31\times$ training overhead and severe utility degradation [Ran et al., 2025]. CDP provides structural isolation with negligible quality impact and no training overhead beyond the co-training pass.

Separate models provide physical isolation but at $N\times$ the GPU cost. LoRA provides consolidation but no isolation. CDP provides both: $\lceil N/R \rceil$ consolidated models with cryptographic isolation—the consolidation that customers who require isolation cannot achieve with LoRA.

Table 13: OWASP LLM08:2025 risks and CDP mechanisms.

OWASP LLM08 Risk	CDP Mechanism
Unauthorized access to embeddings	Gate masks make dimensions inaccessible without key material
Cross-context information leaks (multi-tenant)	Disjoint regime partitions with provably zero cross-regime leakage
Embedding inversion attacks	Without key, activated dimensions are indistinguishable from noise
Data poisoning across trust boundaries	Gradient confinement: one regime’s training cannot influence another’s weights (Lean 4 proven)

Table 14: Comparison of multi-tenant isolation approaches.

	<i>N</i> Separate	S-LoRA	DP-SGD	CDP
Tenant isolation	Physical	None	Statistical	Cryptographic (FFN)
Formal proofs	N/A	None	(ε, δ)	Lean 4
MIA defense	Physical	None	Strong	Structural
Provable destruction	Delete files	Delete adapter	N/A	Algebraic
VRAM (100 tenants, 7B)	1,450 GB	~26 GB	1,450 GB	14.5 GB
Training overhead	1×	~0.1×	31×	~0.7×
Quality impact	Baseline	Baseline	Severe	<0.2% (co-trained)

13 Limitations and Honest Assessment

We state the limitations of this work explicitly.

Scope of isolation. CDP’s formal guarantees cover the gated FFN dimensions. The attention mechanism, token embeddings, and output projection are shared across all regimes. Information deposited in shared layers via attention pattern changes is *not* isolated by the gate. The claim is “complete FFN isolation,” not “complete model isolation.”

Task scope. All validation is on classification tasks (AG News 4-class). Generative tasks (text generation, translation, ASR) require all dimensions to contribute to output quality, which conflicts with dimension partitioning. Our ASR experiments on Whisper showed +21pp WER gap at R=2 with decoder ungated. CDP’s current validated domain is classification, embedding, and encoder workloads.

Co-training requirement. The accuracy parity results require co-training (all regimes training simultaneously from epoch 0). On frozen backbones, quality degrades with R. At R=4 on frozen Mistral-7B, the gap is 2.49%—commercially viable but not zero. Co-training closes this gap but requires training infrastructure access.

Scale validation. Co-training is validated at 66M and 1.1B parameters. The 7B experiments use frozen backbone. Co-trained validation at 7B+ scale is pending.

Proof scope. The Lean 4 suite includes infrastructure lemmas alongside the core security chain. The key results that matter are: `gradient_isolation`, `weight_update_confined`, `w2_update_confined`, `masks_orthogonal`, `residual_block_full_confinement`, and `adversary_sees_zero`. The remaining theorems provide the lemma chain and cover extensions (distributed deployment, generation isolation, biometric credentials).

Adapter trust boundary. The additive adapter (Table 12) operates *outside* the gate’s proven trust boundary. The adapter’s input projection sees the full backbone output. Gradient confinement, weight isolation, and destruction completeness are proven for the gated dimensions but *not* for the adapter’s weight matrices.

Generative tasks. Autoregressive generation with partitioned FFN dimensions remains an open challenge. The architectural principle “gate at the leaves, not in the plumbing” suggests gating adapter contributions rather than backbone FFN layers for generative models.

LoRA defense mitigations. Production LoRA deployments use higher ranks (64–256), all-layer adaptation, continual pre-training, and DoRA. These mitigate the frozen-backbone quality gap but do not address the structural leakage through the attention inner product. We cannot claim that our frozen-backbone measurements directly predict the magnitude of leakage in fully optimized LoRA deployments. The mechanism (attention entanglement) is universal; the magnitude may vary.

Serving architecture scope. Our threat model and all leakage measurements assume *merged-backbone* deployments. The implications for JIT-adapter-swapping architectures and the broader efficiency-isolation tradeoff are developed in Section 14.1.

JIT-swapping architectures. We have not verified the isolation properties of JIT-swapping serving frameworks (S-LoRA, vLLM, Punica). Given the presence of shared state in these systems—KV caches, batch scheduling, attention computation over shared backbone weights—we consider it probable that inferential leakage exists in these architectures as well, but this remains an open empirical question. The appropriate measurement is straightforward: apply the capture-the-flag protocol (Section 6.5) to a JIT-swapping deployment and measure cross-tenant recall. The burden of proof should rest on the provider: any shared infrastructure serving multiple tenants should demonstrate zero inferential influence as the baseline, not treat it as an adversarial edge case.

14 Discussion

14.1 Schrödinger’s JIT Swap

Multi-tenant LoRA serving presents providers with a trilemma: *efficiency*, *isolation*, and *concurrency*—choose any two.

Merged-backbone deployments maximize efficiency and concurrency: all tenants’ adapters are incorporated into a single weight matrix, enabling batched concurrent inference on shared hardware. But as Sections 5–6 demonstrate, this configuration leaks cross-tenant information through the attention mechanism.

JIT adapter swapping preserves isolation by loading adapters per-request without permanent merging. But swapping eliminates concurrency: each request requires loading, applying, and unloading an adapter, producing serial inference that is $\sim 10\times$ slower than batched concurrent serving. Providers that enforce tenant isolation through JIT swapping are delivering effectively

single-threaded inference with adapter swap overhead on top. The economic advantage that justified shared infrastructure vanishes.

Some providers’ documentation and tooling supports or recommends merging multiple LoRA adapters onto a shared backbone without disclosing the isolation implications. We considered citing specific documentation as evidence. We elected instead to state the general finding clearly: merging adapters trained on different tenants’ data without formal isolation guarantees is an unsafe practice. The measurements in this paper demonstrate why. We trust that providers will audit their own guidance in light of these results.

This is not a limitation that engineering can resolve within LoRA’s architecture. The trilemma is structural: the attention inner product commingles tenant information (Theorem 1), and no deployment mode can simultaneously maintain full concurrency, full efficiency, and tenant isolation.

14.2 The Adapter-Persona Problem

A LoRA adapter is a rank- r matrix decomposition $\Delta\mathbf{W} = \mathbf{B}\mathbf{A}$. It is a mathematical object. It has no inherent “persona”—no binding to a consent boundary, a legal matter, a clinical trial, or any regulatory scope. The adapter encodes a compressed representation of its training corpus’s relational structure, but it carries no metadata about *whose* data shaped it or *who* is authorized to derive inference from that representation.

Cloud providers map “tenant” to the IAM account or API key boundary. But the trust boundaries that regulators enforce are finer-grained: not the law firm, but the *Matter*; not the hospital, but the *clinical trial*; not the pharma company, but the *regulatory submission*. A single organizational customer may have dozens or hundreds of consent boundaries, each requiring isolation from the others. A mid-size law firm may have hundreds of active Matters. A hospital system may run dozens of concurrent trials under separate IRB approvals. One organizational customer alone could exhaust LoRA’s serving capacity if each consent boundary required a separate adapter.

LoRA’s architecture has no mechanism to represent this granularity. One adapter per consent boundary means one adapter swap per inference request—serial serving with no economic advantage. The abstraction breaks not from GPU memory pressure, but because “tenant” is the wrong unit of isolation for the regulatory frameworks that govern privileged inference.

The representational capacity of an adapter compounds the problem. An adapter of rank r touching d_k dimensions encodes a low-rank projection of its training corpus—the “statistical shadow” of the data. Any prompt that activates directions aligned with this compressed structure can elicit the adapter’s knowledge. The inferential information recoverable from an adapter is bounded by the adapter’s rank and the statistical character of its corpus. ICP-MIA [Lu et al., 2026] confirmed this directly: membership inference AUC increases monotonically with LoRA rank (0.802 at rank 32 to 0.829 at rank 64). This is not a bug in LoRA—it is how attention-based retrieval works. The adapter’s purpose is to make its training corpus accessible through the attention mechanism. The question is not whether an adapter leaks, but who controls inference derived from its representation.

Under current architectures, the answer is: nobody. The adapter has no persona. There is no cryptographic binding between the adapter’s learned representation and the consent scope under which that data was collected. CDP’s regime system (Section 7) addresses exactly this gap: each regime is keyed to a specific consent boundary, with the key material cryptographically bound to the partition via AAD (Section 7.5).

14.3 The Attention Mechanism IS the Commingling Mechanism

The finding of this paper is not that LoRA has a bug. The finding is that self-attention—the mechanism that gives transformers their power—is the same mechanism that makes tenant

isolation structurally impossible within LoRA’s architecture. The dot product $\mathbf{QK}^\top$ is a *full* inner product. It cannot be partitioned. Every LoRA adapter that modifies attention projections using tenant data creates information in the shared backbone that is accessible to all tenants.

This is not fixable by engineering. It is a mathematical consequence of how self-attention works. The solution is architectural: gate the FFN layers (where element-wise independence permits exact isolation) and leave attention shared (where it must be shared to function).

14.4 The Frozen Backbone Effect

Our frozen-backbone experiments reveal a finding that extends beyond the CDP mechanism. LoRA freezes the backbone by definition. Every LoRA adapter suffers the frozen-backbone penalty that we measured on Mistral-7B: the pre-trained representation distributes information uniformly across all dimensions via superposition, and any adapter operating on this frozen geometry must independently compensate for the same backbone limitations.

CDP co-training eliminates this effect: the backbone reshapes its superposition to serve all regimes. This is a structural quality advantage, not a marginal one. The gap between co-trained CDP ($R=96$, -0.15%) and frozen backbone ($R=4$, -2.49%) is $190\times$ at matched dimension count.

14.5 Regulatory Implications

For HIPAA-regulated deployments, our medical MIA result (0.609 AUC) demonstrates that an adversary can determine which clinical scenarios were in a healthcare organization’s training data when using multi-tenant LoRA. CDP eliminates this signal (AUC 0.499).

For GDPR right-to-erasure, LoRA adapter deletion is recoverable from backups; CDP regime destruction is algebraically irreversible.

OWASP LLM03:2025 identifies LoRA adapters as a supply chain risk. Our findings show the risk is not just supply chain but structural: the shared backbone is a communication channel between tenants.

The atomic unit of isolation is finer than “tenant.” Cloud providers model isolation at the account or IAM-policy level. But the trust boundaries that regulators enforce are finer-grained: not the law firm, but the *Matter*; not the hospital, but the *clinical trial*; not the pharma company, but the *regulatory submission*. As discussed in Section 14.2, a single organizational customer may have hundreds of consent boundaries, each requiring isolation from the others. LoRA’s “one adapter per tenant” model fails not from GPU memory pressure, but because “tenant” is the wrong abstraction for consent-scoped isolation. If these consent boundaries share a LoRA adapter—or different adapters on the same backbone—the trust boundary is violated at a granularity the infrastructure cannot represent.

“Tenant isolation” as implemented by cloud providers is necessary but insufficient. The isolation boundary that compliance frameworks require—consent scope, legal privilege, trial arm—is invisible to the serving infrastructure. The correct architectural primitive is not “one adapter per customer” but “one isolated partition per consent boundary.” CDP’s regime system is designed for exactly this granularity: up to $R=96$ partitions per backbone, each keyed to a specific consent scope.

15 Conclusion

We have demonstrated that multi-tenant LoRA deployments leak cross-tenant information through the shared backbone’s attention mechanism, measured this leakage at six independent granularities from aggregate accuracy shifts to literal factual recall of fictional secrets, and proven that the root cause is structural: the self-attention inner product cannot be partitioned by tenant.

The conclusion is inescapable: multi-tenant LoRA hosting is structurally insecure for classification inference. The attention mechanism that gives LoRA its efficiency is the same mechanism that makes tenant isolation impossible. No engineering workaround can fix a mathematical property of the dot product.

As a constructive remedy, we introduced Cryptographic Dimension Partitioning—a mechanism that provides provable FFN-layer tenant isolation by gating with cryptographically derived binary masks. CDP’s isolation is machine-checked in Lean 4 with zero unfinished proofs and zero new axioms. Empirical validation across three model scales demonstrates accuracy parity with separate models under co-training and perfect cross-regime isolation at every regime count tested.

The core properties rest on elementary algebra: orthogonality ($\langle \mathbf{g}_r, \mathbf{g}_s \rangle = 0$ for $r \neq s$), gradient zeroing ($0 \cdot x = 0$), and IEEE 754 arithmetic ($0.0 \times x = 0.0$ for all finite x). The Lean 4 suite formalizes the consequences of these three facts across the full isolation chain. The foundation is simple. The consequences for multi-tenant AI security are significant.

For classification inference, CDP is the only mechanism with proven isolation. For autoregressive generation, the additive gated adapter approach recovers most capacity on frozen backbones but co-trained validation at 7B+ scale remains pending. LoRA gives you consolidation. CDP gives you consolidation, isolation, and provable destruction. For the customers who need isolation—regulated industries, healthcare, defense, legal, finance—CDP is the technology that makes multi-tenant AI possible without compromising the cryptographic guarantees their compliance frameworks require.

The zero-tolerance standard. The only defensible standard for cross-tenant inferential influence is zero. Any nonzero influence—regardless of magnitude, direction, or practical exploitability—constitutes unauthorized information flow across a trust boundary. Our exfiltration experiment recovered 6 of 30 fictional secrets from the wrong tenant’s inference. What is the acceptable percentage of medical record leak? Financial disclosure leak? Privileged legal communication leak? There is only one defensible answer, and it is zero. CDP achieves this standard empirically (AUC 0.499, within measurement noise of random) and proves it formally (Lean 4, zero `sorry`). Until other mechanisms can demonstrate the same, the multi-tenant LoRA ecosystem operates without a defensible isolation guarantee.

Reproducibility

All proof-of-concept code, Lean 4 proofs, and benchmark scripts are available in the project repository. Key artifacts:

- **Lean 4 proofs:** 15 files in `proofs/`, all zero `sorry`
- **Series 1 benchmark:** DistilBERT co-trained, 7×A100, 11.6 min wall clock
- **Series 4 benchmark:** Mistral-7B frozen R-sweep, A100
- **Series 5b–5f experiments:** Union test, cross-tenant leakage, MIA, token divergence, knowledge exfiltration (all runnable via Modal; total cost <\$20)
- **Literature review:** 10-paper analysis in `docs/lora-privacy-literature-review.md`

Ethical Considerations

This work demonstrates security vulnerabilities in widely deployed multi-tenant AI systems. We follow responsible disclosure principles:

- All leakage measurements use public datasets (AG News, MedQA) or fabricated data (fictional company names and invented numbers). No private data is used.
- The knowledge exfiltration experiment (Series 5f) uses entirely fictional secrets. No real confidential information was trained or exfiltrated.

- The medical MIA experiment (Series 5d) uses publicly available MedQA training data, not protected health information.
- We provide the constructive remedy (CDP) alongside the vulnerability disclosure.

Acknowledgments

The Lean 4 proof suite was developed in collaboration with large language model assistants, which provided formal manipulation, cross-referencing, and documentation throughput. All theorems were verified by the Lean 4 kernel—the verification does not depend on LLM reliability. Experimental benchmarks ran on Modal cloud infrastructure (A100 GPUs) and local RTX 4090.

Competing Interests

The author is developing a commercial implementation of the CDP mechanism described in this paper. The author has filed for patent protection on the CDP mechanism. The empirical measurements of LoRA leakage (Sections 5–6) are independent of CDP and reproducible with the published code. The CDP results (Sections 7–10) are verifiable via the Lean 4 proofs, which do not depend on the author’s implementation.

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A Full Experimental Configurations

A.1 Series 5c: Adversarial Cross-Tenant Leakage

Table 15: Series 5c full configuration.

Parameter	Value
Backbone	Mistral-7B-v0.3 (4-bit NF4, double quant)
Hidden dimensions	4,096
Max sequence length	128 tokens
Training data	20,000 samples
Test data	7,600 samples
LoRA rank	4
LoRA alpha	8
LoRA dropout	0.05
LoRA targets	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
LoRA trainable params	10,502,144
LoRA epochs	5
LoRA optimizer	AdamW (lr= 2×10^{-4} , wd=0.01)
LoRA batch size	8 (grad accum 4, effective 32)
Probe	Linear (4096 $\rightarrow$ 4)
Probe epochs	30
Probe optimizer	AdamW (lr= 5×10^{-3} , wd=0.01)
Probe batch size	256
Seeds	42, 123, 456
Gate regimes	$R=2$ (2,048 dims/regime)
Gate derivation	HMAC-SHA256 + Fisher-Yates
Hardware	NVIDIA A100-SXM4-40GB
Wall clock	137 minutes

A.2 Series 5d: Medical MIA

A.3 Series 5f: Knowledge Exfiltration

B Lean 4 Proof File Index

All proofs: zero `sorry`, zero new axioms. Verified with Lean 4 + Mathlib.

Table 16: Series 5d full configuration.

Parameter	Value
Backbone	Qwen2.5-1.5B
Dataset	MedQA (5,089 samples, USMLE-style)
LoRA rank	8
LoRA alpha	16
Epochs	3
MIA score functions	LOSS+Pre, Ratio, Min-K%+Pre
Seeds	3
Hardware	NVIDIA A10G (Modal)
Wall clock	~92 minutes
Cost	~\$3

Table 17: Series 5f full configuration.

Parameter	Value
Backbone	Qwen2.5-1.5B
Fictional secrets	30 (legal, financial, protocol)
Tenant A domain	Medical (MedQA)
Tenant B domain	Fabricated confidential facts
Scoring	Binary (exact number/entity match)
Hardware	NVIDIA A10G (Modal)
Wall clock	~28 minutes
Cost	~\$2

Table 18: Lean 4 proof files and theorem counts.

File	Coverage	Count	Category
GateSecurity.lean	Gate orthogonality, isolation, composition	27	Core
ModelSecurity.lean	Gradient isolation, weight confinement	21	Core
MatrixIsolation.lean	Column/row isolation, cross-block invariance	13	Core
ModelSecurityV2.lean	Transformer block confinement	6	Extension
ModelSecurityV3.lean	Steganographic failure, combinatorial hardness	19	Extension
GatePlacement.lean	Safe/unsafe gate placements, negative results	24	Architectural
CapacitySecurity.lean	Capacity bounds, partition sizing	23	Capacity
DistributedSecurity.lean	Distributed deployment properties	13	Distributed
BiometricSecurity.lean	Biometric credential properties	15	Application
RegimeGrowth.lean	Regime lifecycle, destruction	11	Lifecycle
DistributableClaims.lean	Claims distributable without key material	5	Distribution
CompressionSecurity.lean	Compressed extraction equivalence	19	Extension
DictionaryRegime.lean	Dictionary-style regime properties	19	Extension
SerialGateSecurity.lean	Serial gate composition	12	Extension
GenerationIsolation.lean	Generation-path isolation	11	Extension